Examiner reports

Q1.

Almost every student selected the correct response to this question. There was no significant pattern in the incorrect responses.

Q3.

Part (a) of this question was usually answered well. There were many correct solutions to part (b) but a few students tried to use the tensions found in part (a) forgetting that there was now an additional particle at B. A significant number of students did not include g in their calculations involving the particle of mass m kg and hence did not take the moment of a force.

Q4.

Many candidates completed this question correctly. A common error occurring in part (a) was that some candidates in their diagram did not show that the two vertical upward reactions on the plank were different, or showed the two vertical gravitational forces acting at the same point. There were some interesting comments made in part (c), but most responses were correct.

Q5.

Many candidates completed this question correctly. A common error occurred in part (a) where some candidates did not show, on their diagram, that the two vertical reactions on the plank were different. Creative algebra by a number of candidates was seen in part (b), particularly in the insertion of 'g' in the final answer.

Q6.

This question was also answered well by many candidates. A few lost a mark in part (a) by not showing that the two tensions were different, and a number of candidates used incorrect moments in part (b), but spuriously obtained the given answer by approximating $\frac{40g}{1.9}$ to $21g$.

Q7.

Diagrams in part (a) were mostly good, with occasional errors in positioning the weight. Part (b) was mostly done correctly but there was some inaccuracy in manipulating simple equations to find the tensions. Most gave appropriate responses in part (c).

Q8.

Many candidates were handicapped by the lack of a clear labelled diagram and so their moment equations were often incorrect. A few candidates forgot to include the g terms.

Q9.

Most candidates were successful in this question. The usual error occurred when candidates took moments about one end, normally the end on the land. They did not consider that the reaction vertically upwards on the rod was $65g$ N, acting at the point where the rod was potentially pivoting.

Q10.

Most candidates did very well on this question, although there were a few who did not seem to be able to attempt this type of problem.

Q11.

There were some very good answers to this question, but a number of areas of difficulty were evident on some scripts. In part (a) a common error was to show the tension acting in an upward direction. Another error that appeared was to omit the normal reaction force. The vast majority of the candidates were able to produce the correct answer in part (b). Some candidates not find the tension directly, but found the reaction force first and then used this to find the tension. The fact that the tension was given helped some candidates, but there were a few solutions where not enough working was shown.

In part (c), one of the most common errors was to omit g from the moment equation, so that one of the terms was the product of a distance and a mass, rather than a distance and a force. This often had an implication for the next stage when the candidates were finding the reaction force. Interestingly, some candidates took moments about the end of the rod, so that having an incorrect mass did not have an impact on their answer. Part (d) was generally done well.

Q12.

This question was generally found to be difficult and many of the weaker candidates did not score many marks on this question. Many candidates produced the answer given in the question, but did not have satisfactory working to support their answer. Those who completed part (a)(i) correctly often went on to do well with part (a)(ii). Part (b) was found to be very difficult, with very few convincing explanations.

Q13.

While there were a number of good solutions to this question, there were a number of difficulties that seemed to originate from not realizing that the reaction force at A acted downwards. This was a particular problem in parts (a) and (c). The printed answer seemed to help quite a lot of the candidates in part (b).

In part (d), the solutions tended to be either very good or very poor. Candidates who made a good start were normally able to find both reactions correctly. However, some hindered themselves by taking moments about points other than A and B or by taking moments, but not expressing these as part of an equation.

Q14.

There were some candidates who did not know how to approach the question. Some candidates produced an equation for vertical equilibrium, but did not take moments correctly, often not multiplying the reaction forces by a distance.